

CARBOXYLATED NBR: PATENT LANDSCAPE AND SYNTHESIS-RELEVANT PROCESS FEATURES

1. ABSTRACT

This report reviews patent evidence relevant to carboxylated nitrile butadiene rubber (NBR), with emphasis on how carboxylated or acid-modified nitrile rubbers are used and processed in heat-activable adhesive systems. The primary patent in the manifest, US20100147462A1, does not disclose a standalone polymerization recipe for making the carboxylated NBR itself; instead, it describes a heat-activable adhesive tape for flexible printed circuit boards based on an acid- or acid anhydride-modified acrylonitrile-butadiene copolymer combined with an epoxy resin. The patent is nevertheless highly relevant because it explicitly identifies the target elastomer class, states the acrylonitrile content range, and shows how carboxylated NBR is formulated to achieve heat activation, solvent processability, high-temperature flow control, and solder-bath resistance.

The disclosed formulation strategy centers on using the acid-modified nitrile rubber as both elastomer and reactive component, eliminating the need for an additional nonpolymer hardener. The weight ratio of modified NBR to epoxy resin is kept above 1.5, which is presented as important for maintaining flexibility while enabling crosslinking during hot lamination and subsequent bonding at elevated temperature. The patent also indicates that the nitrile rubber may be at least partly hydrogenated and that the acrylonitrile content can range from 15% to 50 wt%. Example data show solution coating from butanone, lamination at 100°C, hot pressing at 200°C and 1.3 MPa for 1 hour, and solder-bath testing at 288°C for 10 seconds.

Overall, the patent landscape in this manifest is formulation- and process-oriented rather than monomer-feed-oriented. The key trend is the use of carboxylated NBR as a reactive, high-viscosity elastomer in epoxy-containing adhesive systems for electronics, where thermal stability, low flow, and solvent-based coating are critical. For synthesis planning, the evidence supports selecting an acrylonitrile-rich NBR backbone, introducing acid or acid anhydride functionality, and formulating with epoxy resin at elastomer-rich ratios to obtain heat-activable, crosslinkable materials.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US20100147462A1
- Patent Title: ... strip that can be activated by heat and is based on carboxylated nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2010
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses a heat-activable adhesive tape formulation based on an acid- or acid anhydride-modified acrylonitrile-butadiene copolymer combined with an epoxy resin. The carboxylated NBR is used as a reactive elastomer in a solvent-coated adhesive layer that crosslinks under heat without any additional nonpolymer hardener. - Weight ratio of components a/b: greater than 1.5 — Claims 10, 17 - Acrylonitrile content of acrylonitrile-butadiene copolymer: 15% to 50% by weight — Claim 12 - Lamination temperature: 100 °C — Example bonding procedure - Hot pressing temperature: 200 °C — Example bonding procedure - Pressure: 1.3 MPa — Example bonding procedure - Hot pressing time: 1 hour — Example bonding procedure - Static peel test temperature: 70 °C — Temperature stability test - Static peel load: 500 g — Temperature stability test - Solder bath temperature: 288 °C — Solder bath resistance test - Solder bath exposure time: 10 seconds — Solder bath resistance test - T-peel test rate: 50 mm/min — Test methods - T-peel test angle: 180° — Test methods - T-peel test conditions: 20 °C and 50% relative humidity — Test methods - Coat thickness: 25 µm — Example 1 - Example 1 composition: 80 parts by weight Nipol NX 775 and 20 parts by weight Bakelite EPR 166 — Example 1 - Example 2 composition: 80 parts by weight Nipol NX 775, 20 parts by weight Piccolyte A 125, and Bakelite EPR 166 from Example 1 — Example 2 - Example 3 composition: 75 parts by weight Breon N41H80 and 25 parts by weight Bakelite EPR 166 — Example 3 - Example 4 composition: 70 parts by weight Breon, 25 parts by weight EPR 166, and 5 parts by weight Dyhard 100-S — Example 4 - Breon N41H80 acrylonitrile content: 41% by weight — Example 3 - Breon N41H80 Mooney viscosity ML 1+4 at 100 °C: 72 to 88 — Example 3 - Nipol NX 775 acrylonitrile content: 26% by weight — Example 1 - Nipol NX 775 acid modification: 7% by weight — Example 1 - Bakelite EPR 166 epoxide equivalent: 184 — Example 1

Relevant Experimental Evidence - Example 1: Acid-modified nitrile rubber with epoxy resin gave a coated adhesive layer of 25 µm and served as the main inventive formulation. - Example 2: Addition of polyterpene tackifier to the Example 1 formulation was demonstrated without changing the basic process. - Example 3: Unmodified nitrile rubber with epoxy resin showed much lower bond strength, highlighting the importance of acid modification. - Example 4: A comparative formulation with dicyandiamide hardener was evaluated and used in bonding tests, showing the

role of hardener-free inventive systems.

Technical Relevance This patent is directly relevant because it explicitly identifies acid-modified acrylonitrile-butadiene copolymers as the key elastomer class for heat-activable adhesive systems. It provides practical formulation and bonding conditions that show how carboxylated NBR is used in a reactive, thermally stable application rather than as a standalone polymerization example.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- No qualifying primary patents were identified under the configured target material relevance criteria. Cross-patent quantitative trends were therefore not generated.

4. CONCLUSION

The available evidence supports carboxylated NBR as a reactive elastomer for heat-activable, epoxy-containing adhesive systems rather than a detailed monomer-by-monomer synthesis disclosure. The most important evidence-based parameters are an acrylonitrile content of 15% to 50 wt%, acid or acid anhydride modification, and an elastomer-to-epoxy weight ratio greater than 1.5, with no additional nonpolymer hardener. For process implementation, solvent coating from butanone, lamination at 100 °C, and hot pressing at 200 °C and 1.3 MPa for 1 hour are the clearest disclosed operating conditions. The patent indicates that these choices improve bond strength, thermal stability, and solder-bath resistance while preserving processability.

5. REFERENCES

- US20100147462A1 | ... strip that can be activated by heat and is based on carboxylated nitrile rubber | Not disclosed | US | 2010 | <https://patents.google.com/patent/US20100147462A1/en>